IRYOTEK HEALTHCARE MANAGEMENT ECOSYSTEM

Comprehensive Entity-Relationship (ER) Relational Database Architecture Diagram • MySQL 8.0 InnoDB

DATABASE: hospital

22 Relational Entities • 100% Referential Integrity

specialist

PK	**id**	INT (PK)
•	specialist_name	VARCHAR(100)

hospital

PK	**id**	INT (PK)
•	name	VARCHAR(200)
•	city	VARCHAR(100)
•	state	VARCHAR(100)
•	phone	VARCHAR(20)

belongs to (N : 1)

patient_profile

PK	**id**	INT (PK)
FK	**user_id**	INT (FK)
•	date_of_birth	VARCHAR(50)
•	gender	VARCHAR(20)
•	blood_group	VARCHAR(10)
•	emergency_phone	VARCHAR(20)
•	emergency_contact	VARCHAR(100)

ai_differential_diagnosis

PK	**id**	INT (PK)
FK	**user_id**	INT (FK)
FK	**appointment_id**	INT (FK)
•	urgency_level	VARCHAR(50) ['RED'/'YEL'/'GRN']
•	symptoms	TEXT
•	diagnoses_json	JSON/TEXT
•	clinical_notes	TEXT

doctor

PK	**id**	INT (PK)
FK	**hospitalId**	INT (FK)
•	fullName	VARCHAR(100)
•	qualification	VARCHAR(100)
•	specialist	VARCHAR(100)
•	email	VARCHAR(100) [UK]
•	phone	VARCHAR(20)
•	timing	VARCHAR(100)

written by (N : 1)

profile of (1 : 1)

assessed for (N : 1)

evaluated on (N : 1)

attends (N : 1)

issued to (N : 1)

prescription

PK	**id**	INT (PK)
FK	**appointment_id**	INT (FK)
FK	**user_id**	INT (FK)
FK	**doctor_id**	INT (FK)
•	status	VARCHAR(50)
•	remarks	TEXT
•	approved_at	TIMESTAMP

fulfills (N : 1)

pharmacy_dispensing_request

PK	**id**	INT (PK)
FK	**prescription_id**	INT (FK)
FK	**medicine_id**	INT (FK)
FK	**patient_id**	INT (FK)
FK	**doctor_id**	INT (FK)
•	calculated_quantity	INT
•	quantity_dispensed	INT
•	quantity_remaining	INT
•	status	VARCHAR(50)
•	idempotency_key	VARCHAR(100) [UK]

audits (N : 1)

pharmacy_audit_log

PK	**id**	INT (PK)
FK	**dispensing_request_id**	INT (FK)
FK	**medicine_id**	INT (FK)
•	pharmacist_id	INT
•	action	VARCHAR(50)
•	quantity_dispensed	INT
•	previous_stock	INT
•	new_stock	INT
•	timestamp	TIMESTAMP

dispenses (N : 1)

patient_vitals (time-series)

PK	**id**	INT (PK)
FK	**user_id**	INT (FK)
•	recorded_date	VARCHAR(50)
•	heart_rate	INT (bpm)
•	bp_systolic	INT (mmHg)
•	bp_diastolic	INT (mmHg)
•	oxygen_saturation	INT (% SpO2)
•	blood_glucose	INT (mg/dL)
•	temperature	DOUBLE (°F)
•	created_at	TIMESTAMP

logged for (N : 1)

user_details (patient)

PK	**id**	INT (PK)
•	full_name	VARCHAR(100)
•	email	VARCHAR(100) [UK]
•	password	VARCHAR(255) [BCrypt]

generated for (1 : 1)

appointment

PK	**id**	INT (PK)
FK	**userId**	INT (FK)
FK	**doctorId**	INT (FK)
•	tokenNo	INT
•	appointmentDate	VARCHAR(50)
•	appointmentTime	VARCHAR(50)
•	diseases	VARCHAR(255)
•	status	VARCHAR(50)
•	fullName	VARCHAR(100)
•	phone	VARCHAR(20)

booked by (N : 1)

contains (N : 1)

medicine

PK	**id**	INT (PK)
•	name	VARCHAR(200)
•	category	VARCHAR(100)
•	dosage_form	VARCHAR(100)
•	price	DOUBLE
•	stock	INT
•	description	TEXT

prescription_medicine

PK	**id**	INT (PK)
FK	**prescription_id**	INT (FK)
FK	**medicine_id**	INT (FK)
•	medicine_name	VARCHAR(200)
•	strength	VARCHAR(100)
•	dosage_form	VARCHAR(100)
•	units_per_dose	DOUBLE
•	frequency	VARCHAR(100)
•	duration_days	INT
•	calculated_qty	INT

prescribes (N : 1)

references (N : 1)

patient_medical_history

PK	**id**	INT (PK)
FK	**user_id**	INT (FK)
•	category	VARCHAR(50) ['ALLERGY'/'SURGERY']
•	title	VARCHAR(200)
•	diagnosed_year	VARCHAR(50)
•	description	TEXT

history of (N : 1)

uploaded by (N : 1)

policy for (1 : 1)

held by (1 : 1)

patient_insurance

PK	**id**	INT (PK)
FK	**user_id**	INT (FK)
•	provider_name	VARCHAR(150)
•	policy_number	VARCHAR(100)
•	sum_insured	DOUBLE (e.g. 500k)
•	deductible_total	DOUBLE
•	status	VARCHAR(50) [ACTIVE]
•	expiry_date	VARCHAR(50)

filed by (N : 1)

delivery for (N : 1)

placed by (N : 1)

cart

PK	**id**	INT (PK)
FK	**user_id**	INT [UK] (FK)
•	updated_at	TIMESTAMP

items in (N : 1)

cart_item

PK	**id**	INT (PK)
FK	**cart_id**	INT (FK)
FK	**medicine_id**	INT (FK)
•	quantity	INT
•	dosage_instructions	VARCHAR(255)
•	source	VARCHAR(50) ['RX'/'SHOP']
•	prescription_id	INT

medicine_order

PK	**id**	INT (PK)
•	order_number	VARCHAR(100) [UK]
FK	**user_id**	INT (FK)
FK	**address_id**	INT (FK)
•	total_amount	DOUBLE
•	payment_method	VARCHAR(50)
•	payment_status	VARCHAR(50)
•	order_status	VARCHAR(50)
•	created_at	TIMESTAMP

order lines (N : 1)

medicine_order_item

PK	**id**	INT (PK)
FK	**order_id**	INT (FK)
FK	**medicine_id**	INT (FK)
•	medicine_name	VARCHAR(200)
•	price	DOUBLE
•	quantity	INT
•	subtotal	DOUBLE

contains (N : 1)

patient_document

PK	**id**	INT (PK)
FK	**user_id**	INT (FK)
•	document_name	VARCHAR(255)
•	document_type	VARCHAR(100)
•	file_path	VARCHAR(500)
•	uploaded_at	TIMESTAMP

claimed under (N : 1)

user_address

PK	**id**	INT (PK)
FK	**user_id**	INT (FK)
•	full_name	VARCHAR(100)
•	phone	VARCHAR(20)
•	street_address	VARCHAR(500)

shipped to (N : 1)

insurance_claim

FK	**insurance_id**	INT (FK)
•	treatment_desc	VARCHAR(500)
•	total_billed_amount	DOUBLE
•	covered_amount	DOUBLE
•	patient_copay	DOUBLE
•	claim_status	VARCHAR(50) ['APPROVED'/'REVIEW']

FUNCTIONAL DOMAIN LEGEND & CARDINALITY:

Core Clinical & Appointments · Smart Prescription & Dispensing · Pharmacy Orders & Logistics · Patient EHR, Vitals & AI Triage · Health Insurance & Claims

KEYS: [PK] Primary Key (Auto-Increment) [FK] Foreign Key Constraint [UK] Unique Indexed Field CARDINALITY: 1:1 One-to-One | 1:N One-to-Many | Arrow: References Target Entity